



HIVE-FU

Team 14

1969

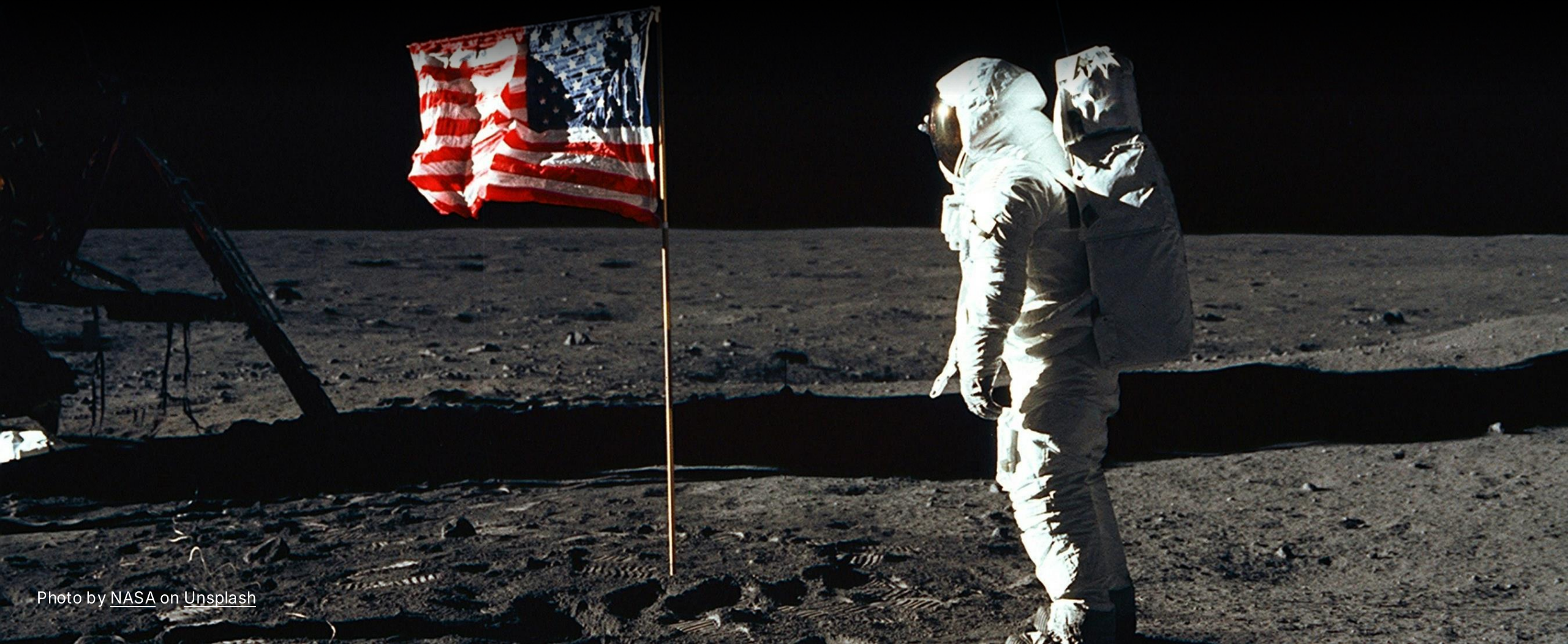
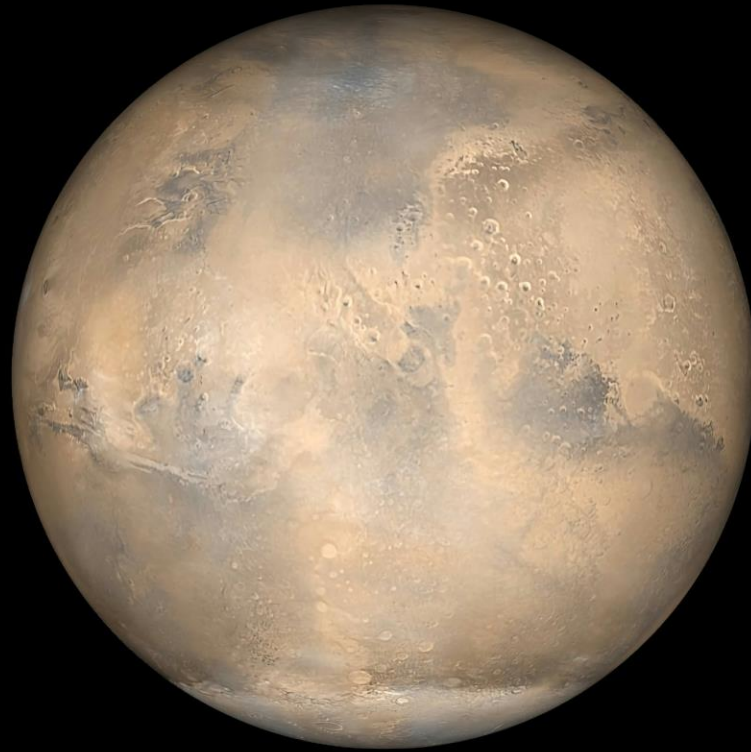
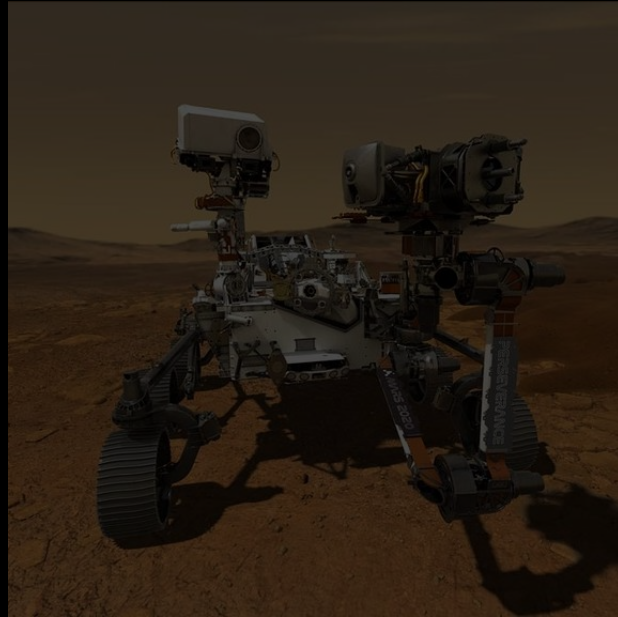


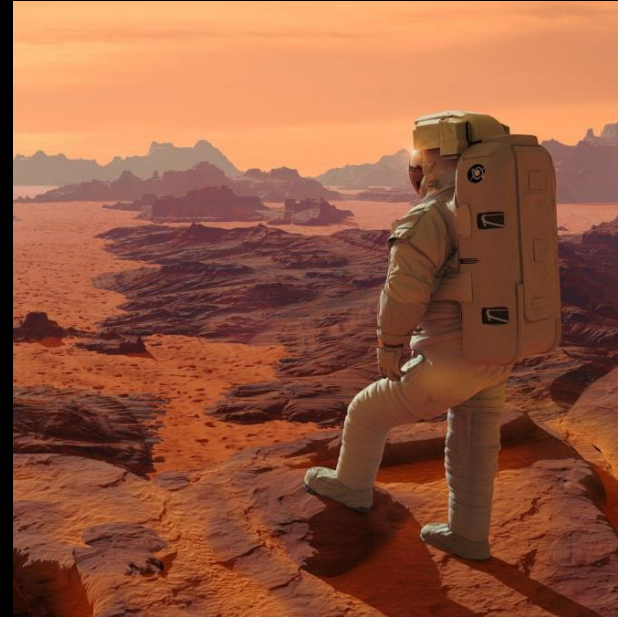
Photo by [NASA](#) on [Unsplash](#)

2025





[1]



[2]

[1] "Meet Perseverance, NASA's Newest Mars Rover," *NASA Jet Propulsion Laboratory (JPL)*, 2021. <https://www.jpl.nasa.gov/edu/resources/teachable-moment/meet-perseverance-nasas-newest-mars-rover/>

[2] J. K. N. Reporter, "Humans Could Land on Mars in 5 to 10 Years, if Elon Musk Has His Way," *Newsweek*, Dec. 28, 2021. <https://www.newsweek.com/humans-could-land-mars-5-10-years-if-elon-musk-has-his-way-1663860>



**"Medical care is the
barrier."**

Dr Thais Russomano

MD, MSc, PhD, FRSA

Doctor specialising in space medicine and physiology

Consulted in projects for ESA and NASA



News & Events ▾ Multin

3 MIN READ

How do Dangerous Blood Clots Form in Space? New Insights from a Valve Model May Provide Information to Protect Astronauts

[3]

Risk of Kidney Stone Formation



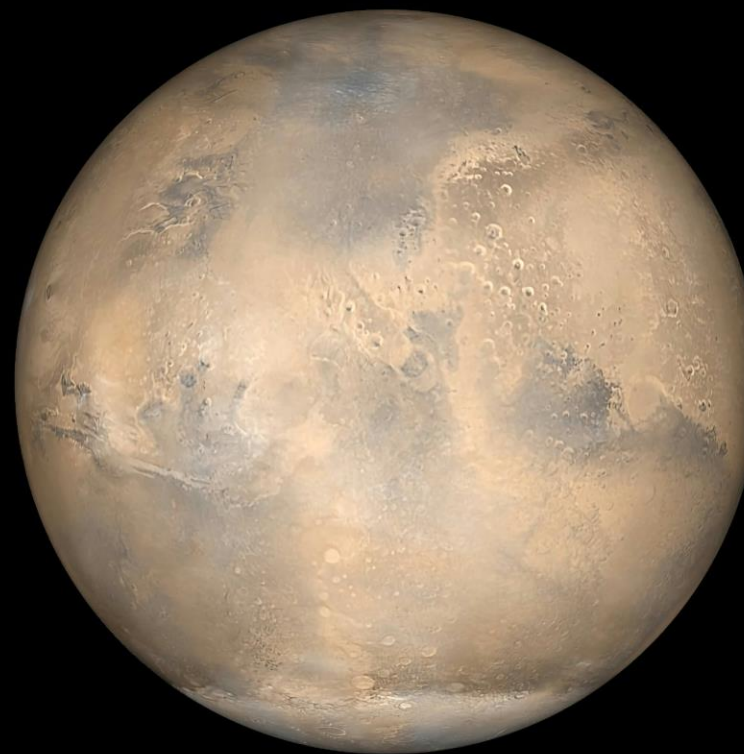
NASA astronaut Mark Vande Hei conducts kidney cell research inside the Life Sciences Glovebox aboard the International Space Station. Such work targets maladies that astronauts are at risk of developing during missions, including kidney stones. Credit: NASA

[4]

[1] "How do Dangerous Blood Clots Form in Space? New Insights from a Valve Model May Provide Information to Protect Astronauts - NASA Science," *science.nasa.gov*. <https://science.nasa.gov/science-research/science-enabling-technology/how-do-dangerous-blood-clots-form-in-space/>

[2] "Risk of Kidney Stone Formation - NASA," NASA, Oct. 21, 2024. <https://www.nasa.gov/reference/risk-of-kidney-stone-formation/> (accessed Feb. 28, 2025).

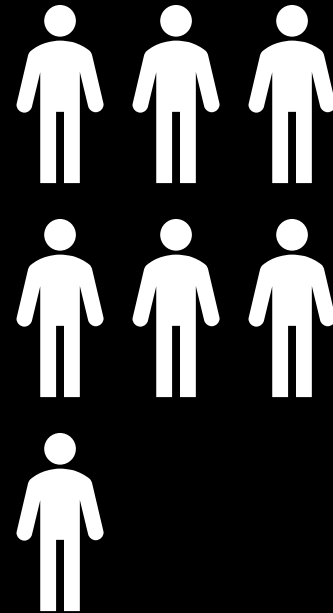
3 years





Mars

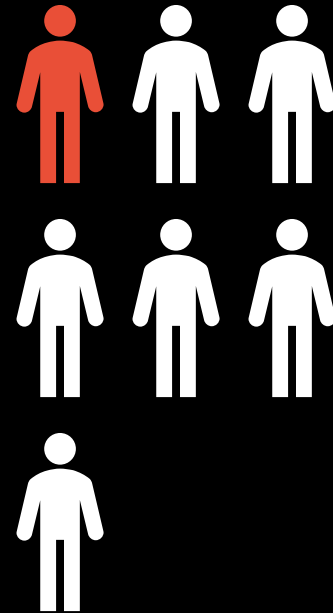
Duration 2.5 years
Crew Size 7





Mars

Duration 2.5 years
Crew Size 7





[6]



[8]



[9]



[10]

Communication Delays

Mars missions will face up to 20 minute delays, one way [7].



20 minutes



Telemedicine is not possible.

[6] "Innovative 3D Telemedicine to Help Keep Astronauts Healthy - NASA," Apr. 08, 2022.

<https://www.nasa.gov/humans-in-space/innovative-3d-telemedicine-to-help-keep-astronauts-healthy/>

[7] NASA, "Hazard: Distance from Earth - NASA," NASA, Aug. 01, 2024. <https://www.nasa.gov/hrp/hazard-distance-from-earth/>



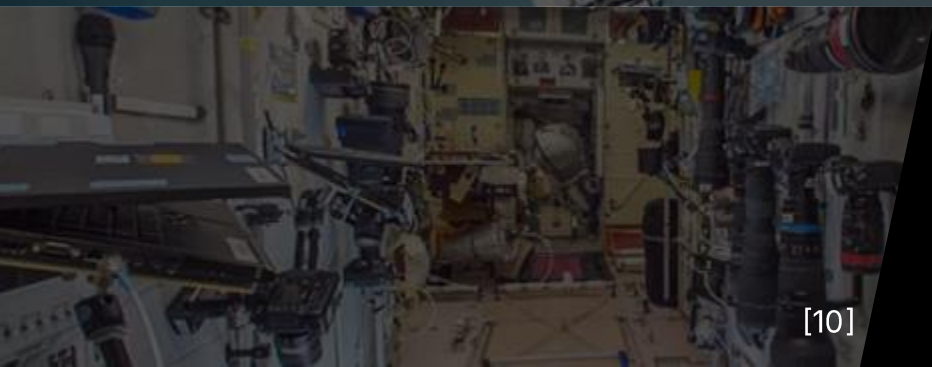
[6]



[8]



[9]



[10]

Medical Emergencies

Medical emergencies must be diagnosed and treated by the crew, without a flight surgeon or reliance on tele-medicine [7].

[8] "Risk of Adverse Health Outcomes and Decrements in Performance due to Inflight Medical Conditions - NASA," NASA, Mar. 20, 2023. <https://www.nasa.gov/directorates/esdmd/hhp/risk-of-adverse-health-outcomes-and-decrements-in-performance-due-to-inflight-medical-conditions/> (accessed Feb. 28, 2025).



[6]



[8]



[9]



[10]

Resupply

Given the distance (53 million – 400 million km) from Earth, resupply is not an option, hence the crew must be completely self-sufficient [7].

[9] "NASA Invites Media to Northrop Grumman's 21st Station Resupply Launch - NASA."
<https://www.nasa.gov/news-release/nasa-invites-media-to-northrop-grumman-21st-station-resupply-launch/>



[6]

Equipment Failure

Should equipment fail, there would be limited access to spare parts or replacement tools [7].



[8]



[9]



[10]

[10]"See 360-degree view inside International Space Station," *euronews*, Jul. 17, 2019.
<https://www.euronews.com/health/2019/07/17/see-360-degree-view-inside-international-space-station>

**Self-sufficiency is
essential for successful
missions to Mars [7].**



**There exists no strategy
for severe medical
emergencies [13][14].**



[12]



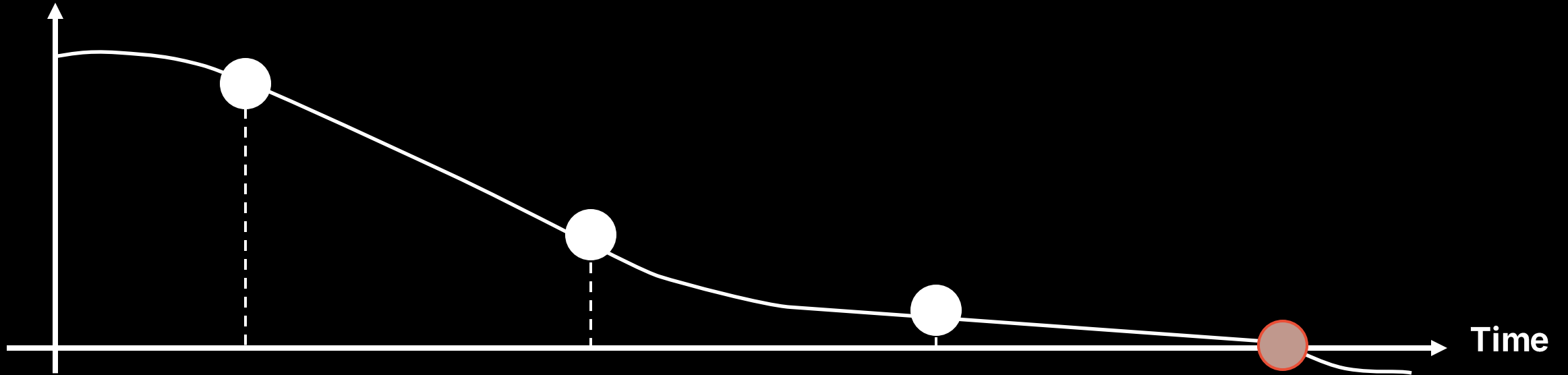
[13]

[12] Wikipedia, "Soyuz MS," Wikipedia, n.d.. [Online]. Available: https://en.wikipedia.org/wiki/Soyuz_MS. [Accessed 18 November 2024].

[13] M. Yates, "Mark Yates charts the development of Apollo's portable life support system," The Chemical Engineer, 18 July 2019. [Online]. Available: <https://www.thechemicalengineer.com/features/first-steps/>. [Accessed 18 November 2024].

[14] H. Hashmatullah, "System-Level Design of a Space Ambulance in Low-Earth Orbit," University of California, Davis, California, 2019.

**Mission
Success**



**Onset of
Symptoms**

Mild pain and
discomfort

**Condition
Worsens**

Pain management
and consultation

**Emergency
Response**

Pain now severe, has
difficulty working

**Mission
Failure**

Incapacitation and
long-term damage



[15]

Valentin Lebedev

Russian Astronaut
Salyut Programme, 1971-1986

**Ultrasound to build a fully
integrated system for diagnosis,
planning and treatment.**

Ultrasound



[16]

Used on Earth for
therapy



[17]

Used on ISS for
imaging



[18]

Multi-functional



[19]

Integration

[16] "HIFU for Prostate Cancer - Saint John's Cancer Institute - Santa Monica, CA," *Urology*, May 30, 2024. <https://www.saintjohnscancer.org/urology/high-intensity-focused-ultrasound-hifu-for-prostate-cancer/> (accessed Feb. 28, 2025).

[17] "NASA Image and Video Library," *NASA Image and Video Library*, 2025. <https://images.nasa.gov/details/iss065e069159> (accessed Feb. 28, 2025).

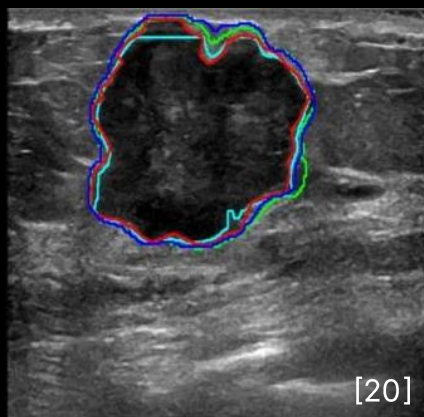
[18] "Affiniti 30," *Philips*, 2025. <https://www.philips.co.uk/healthcare/product/HC795218/affiniti-30-ultrasound-system> (accessed Feb. 28, 2025).

[19] "HIFU for Prostate Cancer - Saint John's Cancer Institute - Santa Monica, CA," *Urology*, May 30, 2024. <https://www.saintjohnscancer.org/urology/high-intensity-focused-ultrasound-hifu-for-prostate-cancer/> (accessed Feb. 28, 2025).

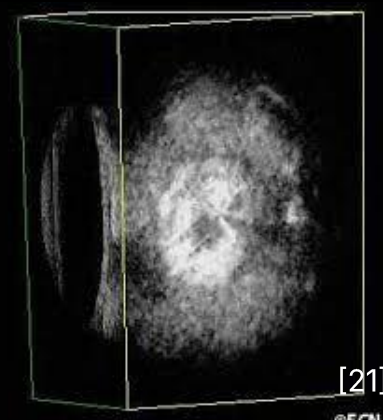
Integration



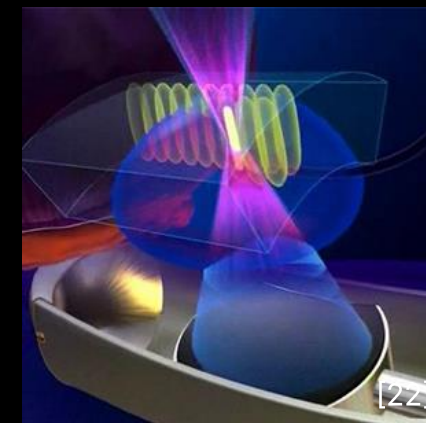
Scan



Identify



Plan



Treat

[20] "How AI and Deep Learning Will Enable Cancer Diagnosis Via Ultrasound," *Imaging Technology News*, Jun. 12, 2018. <https://www.itnonline.com/content/how-ai-and-deep-learning-will-enable-cancer-diagnosis-ultrasound>

[21] NYECC, "Ultrasound Images» New York Eye Cancer Center," *New York Eye Cancer Center*, May 09, 2016. <https://eyecancer.com/eye-cancer/image-galleries/ultrasound-images/>

[22] "HIFU - High Intensity Focused Ultrasound," *Urology in Columbus Ohio, BPH, ED, Prostate Cancer Treatment*, Aug. 16, 2023. <https://www.centralohiourology.com/hifu/> (accessed Feb. 28, 2025).



"A fully integrated solution further enables crew autonomy."

Dr Thais Russomano

MD, MSc, PhD, FRSA

Doctor specialising in space medicine and physiology

Consulted in projects for ESA and NASA



Plan of Embodiment

Design Development



Initial
Ideation

Idea
Sketching

Idea
Selection

Developed Embodiment

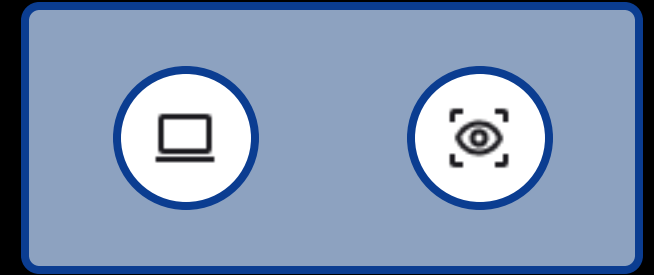


Idea
Refinement

Performance
Evaluation

Expert
Evaluation

Digital Embodiment



Computer-Aided
Design

Render



Prof. Jerome Apt
Former Astronaut



Prof Thais Russomano
*Doctor specialising in
space medicine and
space physiology*



Dr Andrew Cowburn
*Director of Remote
Medicine at Imperial
College London*



Gang Kai Poh
NASA Researcher



Athanasios Viontzos
*Expert on future Mars
missions and Machine
Learning-Supported
Diagnostic Systems*



Dr Sophie Morse
*Expert on therapeutic
ultrasound, PhD in Non-
invasive Surgery and Biopsy*



Matvey Boguslavskiy
*Former Imperial Space
Society President*



Kenton Kwok
*Former Imperial
Physicist, Master Thesis
focused on ultrasound*

Problems

- 1 Crew must handle medical emergencies independently.
- 2 Current ultrasound systems are fragmented and energy-intensive.
- 3 System must meet mass and volume constraints for spaceflight.

How Might We...

 Design Development

 Developed Embodiment

 Digital Embodiment

Enable crew members
to administer
ultrasound treatments?

Enable the use of
multiple ultrasound
modes?

Develop the system to
be modular and field-
repairable?

Develop an assisted
system for diagnosis
and treatment?

Make the system
compact and
lightweight?

Ensure energy-
efficiency for power
constraints?

Design Criteria

● Design Development

● Developed Embodiment

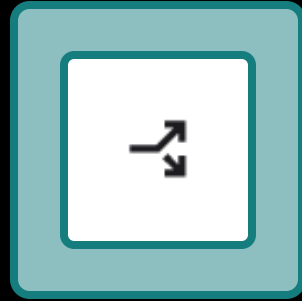
● Digital Embodiment

20%



User Operability

20%



Modality Flexibility

20%



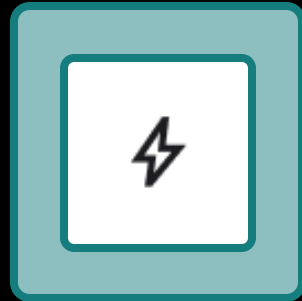
Reliable

15%



Repair Efficiency

15%



Lightweight

10%



Energy Efficiency



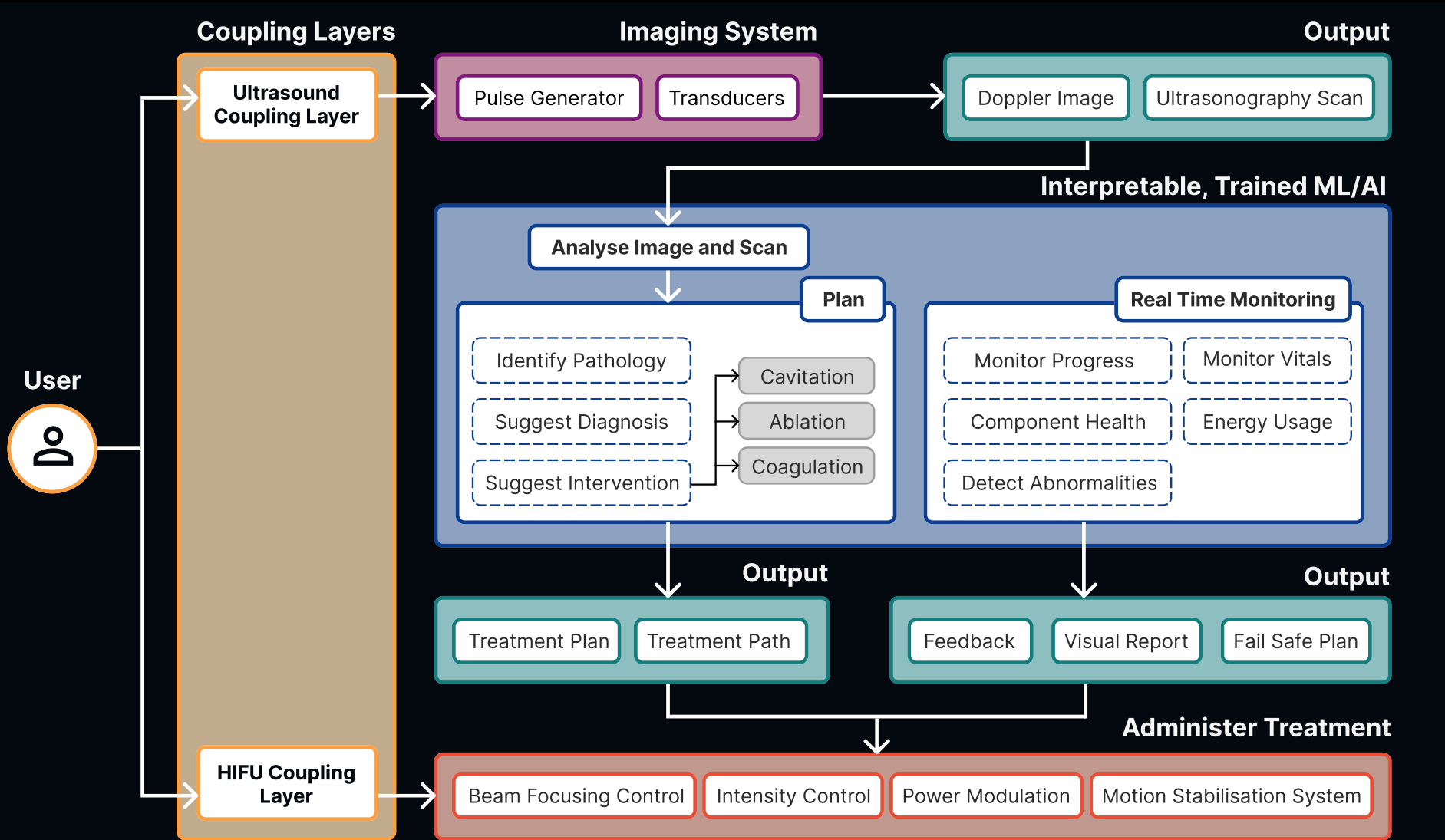
Prof. Jerome Apt
Former Astronaut



Prof Thais Russomano
*Doctor specialising in
space medicine and
space physiology*

System Diagram

- Design Development
- Developed Embodiment
- Digital Embodiment



Athanasios Vlontzos

Expert on future Mars missions and Machine Learning-Supported Diagnostic Systems

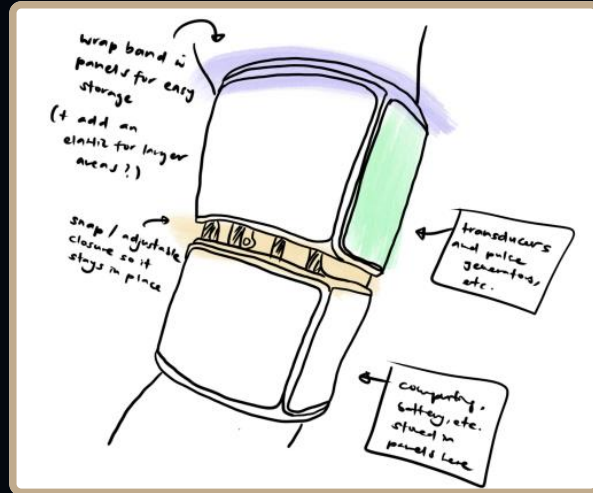


Kenton Kwok

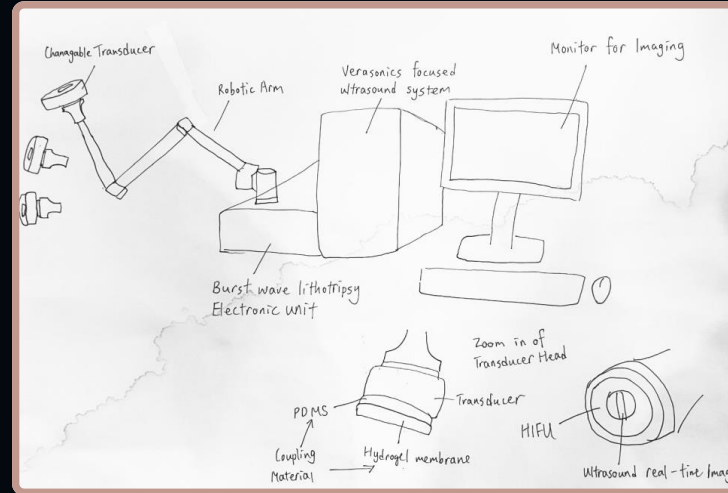
Former Imperial Physicist, Master Thesis focused on ultrasound

Ideation Sketching

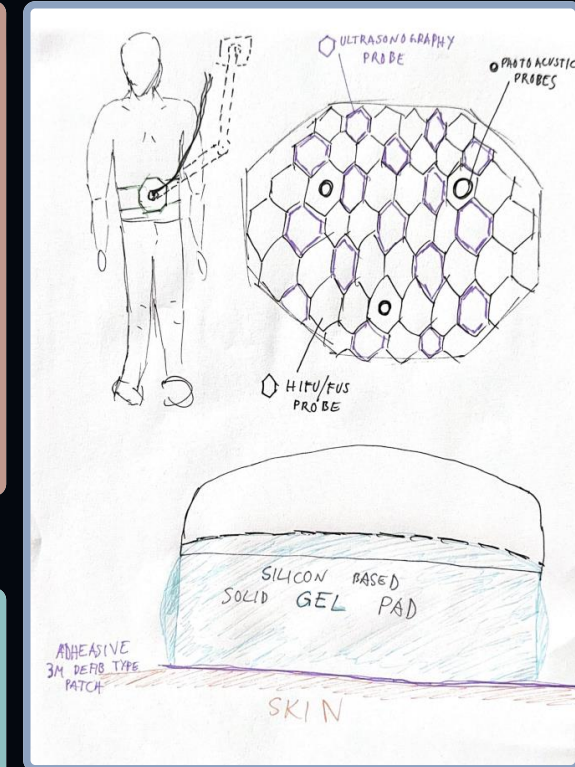
HIFU-Band



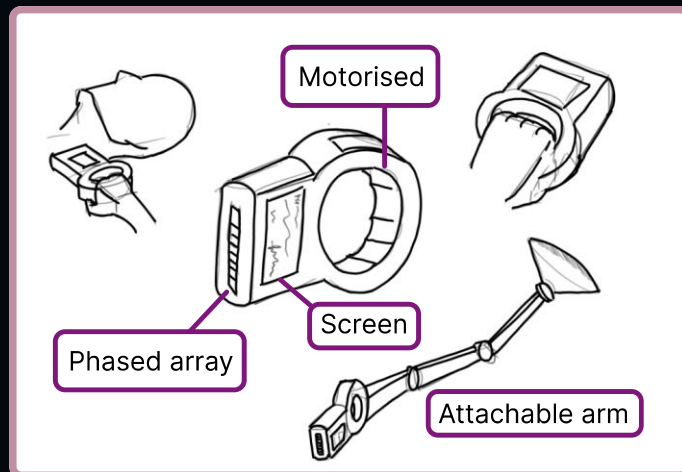
Verasonics



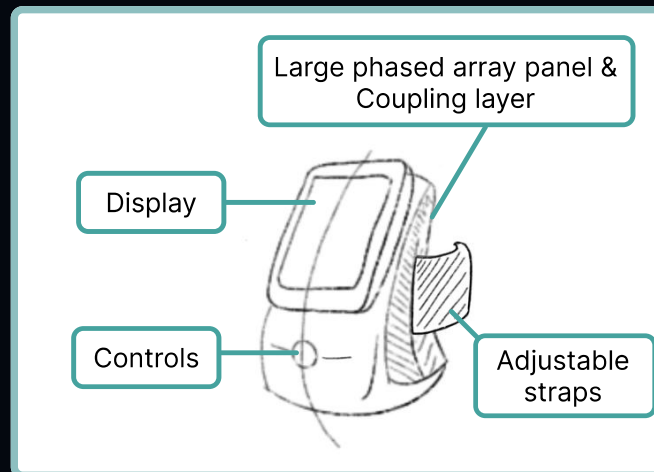
HIVE-FU



SpaceCarer



AstroMed



While many ideas shared overlapping or similar elements, some offered radically different approaches. These concepts are now being analysed and assessed to select the final design.

Idea Selection

Weighted Score

Criteria	Weight	HIFU-Band	Versaonics	HIVE-FU	SpaceCarer	AstroMed
User Operability	20%	4	5	4	5	4
Modality Flexibility	20%	3	4	4	4	4
Reliability	20%	3	4	5	4	3
Repair Efficiency	15%	3	4	4	4	3
Lightweight	15%	5	2	3	4	3
Energy Efficient	10%	4	4	4	4	4
Total	100%	3.6	3.9	4.05	4.2	3.5

Idea Refinement



Athanasios Vlontzos

Expert on future Mars missions and Machine Learning-Supported Diagnostic Systems



Ranked SpaceCarer highest

For modular design, handheld capabilities, which offers versatility.



Ranked HIFU-band lowest

Due to smaller size and static nature which limits its use cases.



Kenton Kwok

Former Imperial Physicist, Master Thesis focused on ultrasound



Ranked HIVE-FU highest

For innovative transducer array and integrated multi-modal imaging.



Ranked AstroMed lowest

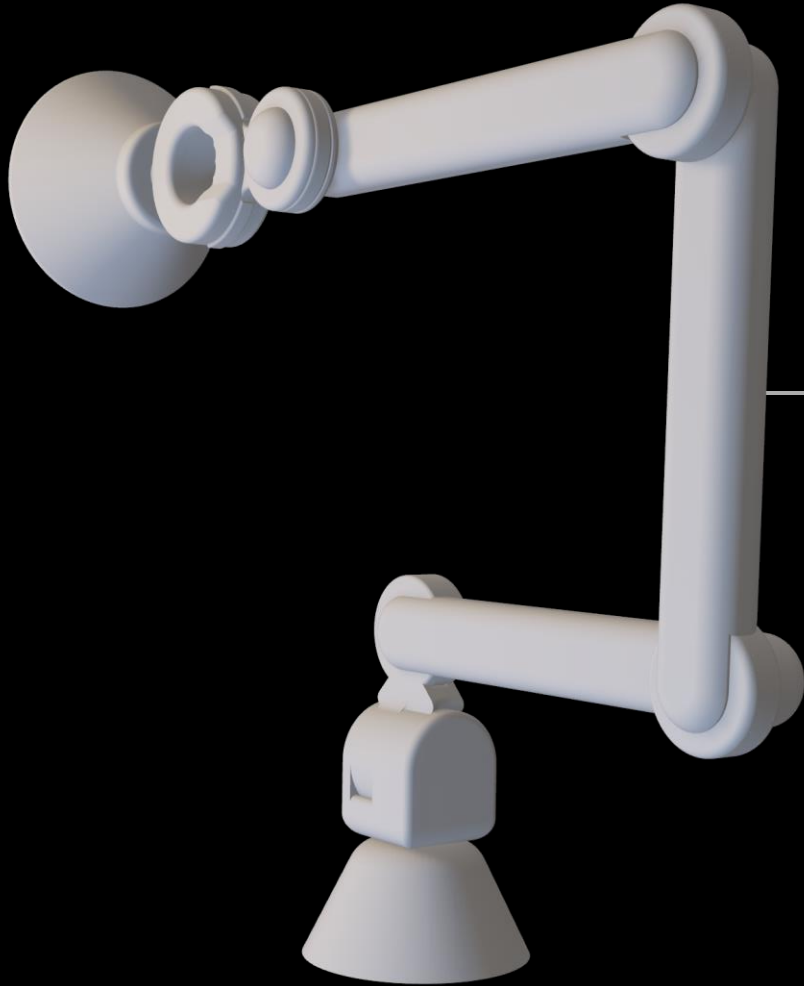
Due to restrictive and bulky design which limits effective imaging and treatment on all body parts.

HIVE-FU



- Design Development
- Developed Embodiment
- Digital Embodiment

- Design Development
- Developed Embodiment
- Digital Embodiment



Motion

6 degree-of-freedom robotic arm
compensates for motion disturbances

Transducers

Concave head features hexagonal tessellation of transducers to enable both ultrasound imaging and HIFU



Human-In-The-Loop

Intuitive user-interface to enable crew members to deliver treatment, monitor progress and enable the fail-safe.



Taavi Mets
Control Engineer
Age: 34 Allergens: Chicken
Weight: 85 kg Medication: N/A
Height: 187 cm Existing Conditions: N/A

Health Log



Health Checkup

T+24: Diagnosis, kidney stone, attention required



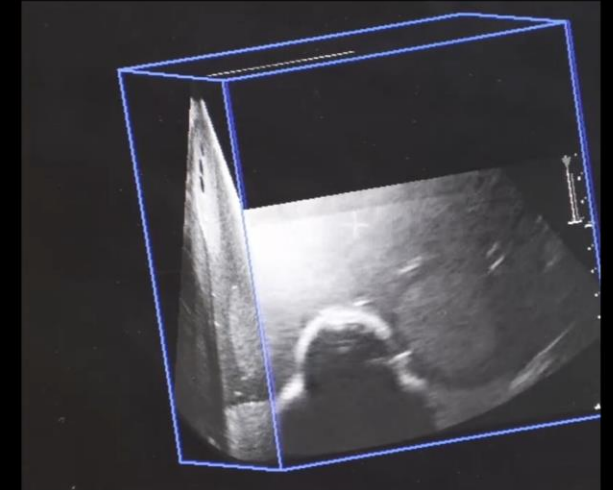
Health Checkup

T+17: Pain in lower abdominals



Health Checkup

T+17: Pain in lower abdominals



Evaluation

Engineering



Human Factors



Feasibility

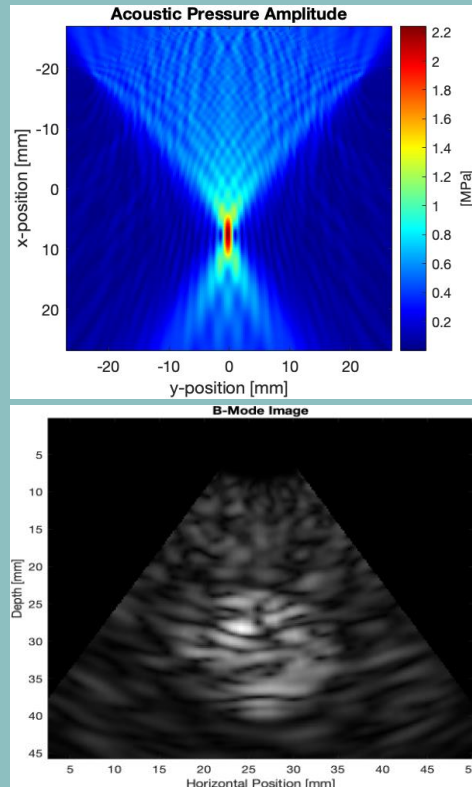


Engineering Analysis

[23] B. Potočník and M. Šavc, "Deeply-Supervised 3D Convolutional Neural Networks for Automated Ovary and Follicle Detection from Ultrasound Volumes," *Applied Sciences*, vol. 12, no. 3, p. 1246, Jan. 2022, doi: <https://doi.org/10.3390/app12031246>.

Transducer setup

- **Dual Functionality** of transducers for therapy and imaging

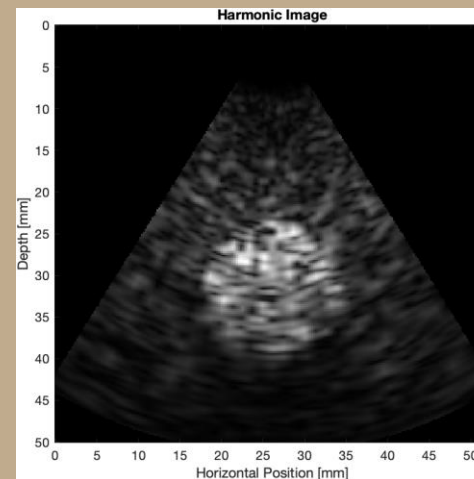


Interference

- **Time-multiplexed** operation for interference elimination

HIFU execution:
1-50ms

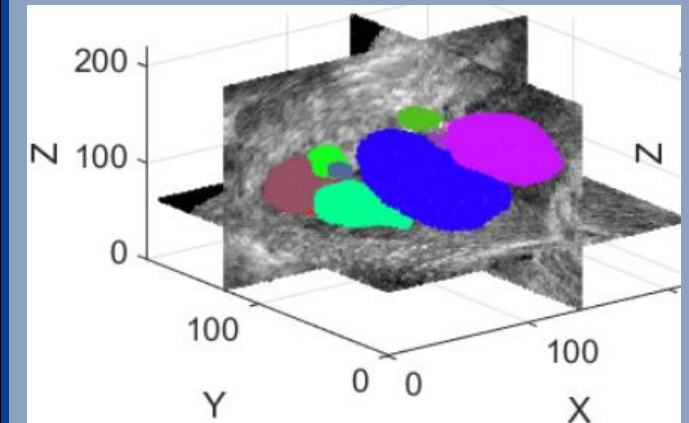
Imaging during cooling:
300-700 μ s



3D Imaging & Detection

- 3D imaging capability with AI-enhanced processing

85% accuracy
improvement 3D vs. 2D



CNN (convolutional
neural network) [23]

Human Factors



Dr Andrew Cowburn
*Director of Remote
Medicine at Imperial
College London*

“In any incident, there’s no one to blame. Like surgery, every participant gives informed consent. With AI assistance, the decision is mutual, and responsibility is shared.”

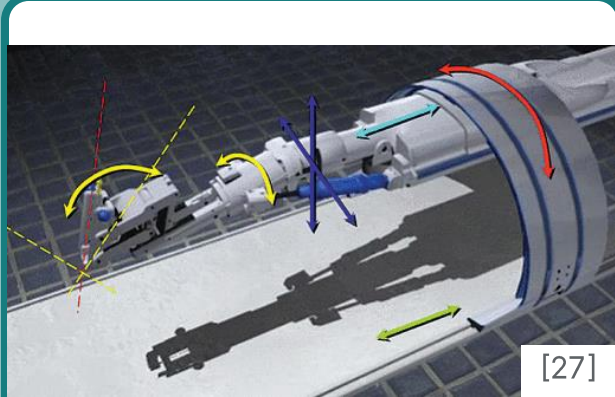


[24] A. Guzman, "Ultrasound Scans in Space Transform Medicine on Earth - NASA," NASA, Jul. 23, 2022. <https://www.nasa.gov/missions/station/ultrasound-scans-in-space-transform-medicine-on-earth/>

[25] E. Howell, "Final Frontier Medicine: Astronauts Use 'Smart Shirt' and Ultrasounds to Monitor Health," *Space.com*, Feb. 09, 2019. <https://www.space.com/43285-astronaut-health-medicine-in-space.html> (accessed Feb. 28, 2025).

[26] Mindo, "The definition of consent - Medical Independent," *Medical Independent*, Oct. 14, 2022. <https://www.medicalindependent.ie/comment/opinion/the-definition-of-consent/> (accessed Feb. 28, 2025).

Feasibility



[27]

OBJECTIVE STRUCTURED CLINICAL EVALUATION (OSCE) OF AN ARTIFICIAL INTELLIGENCE (AI) CLINICAL DECISION SUPPORT SYSTEM (CDSS) TOOL

Human Research Program

February 2025

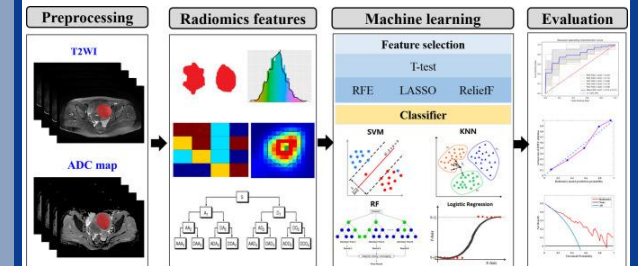
Ariana Nelson¹, Ali Al², Martin Garcia³, Truong Le³, Lynn Boley², Kurt L. Berens², Jay Lemery³

¹University of Texas Medical Branch, Galveston, TX.

²KBR, Inc., Houston, TX.

³NASA Johnson Space Center.

[28]



[29]

[27] A. Gunderman, R. Montayre, A. Ranjan, and Y. Chen, "Review of Robot-Assisted HIFU Therapy," *Sensors*, vol. 23, no. 7, pp. 3707–3707, Apr. 2023, doi: <https://doi.org/10.3390/s23073707>

[28] H. Program *et al.*, "OBJECTIVE STRUCTURED CLINICAL EVALUATION (OSCE) OF AN ARTIFICIAL INTELLIGENCE (AI) CLINICAL DECISION SUPPORT SYSTEM (CDSS) TOOL OBJECTIVE STRUCTURED CLINICAL EVALUATION (OSCE) OF AN ARTIFICIAL INTELLIGENCE (AI) CLINICAL DECISION SUPPORT SYSTEM (CDSS) TOOL 'Expanding the Boundaries of Space Medicine and Technology,'" 2025. Accessed: Feb. 28, 2025. [Online]. Available: <https://ntrs.nasa.gov/api/citations/20250000051/downloads/IWS%202025%20OSCE%20for%20AI%20CDSS.pdf>

[29] Y. Zheng, L. Chen, M. Liu, J. Wu, R. Yu, and F. Lv, "Prediction of Clinical Outcome for High-Intensity Focused Ultrasound Ablation of Uterine Leiomyomas Using Multiparametric MRI Radiomics-Based Machine Learning Model," *Frontiers in Oncology*, vol. 11, Sep. 2021, doi: <https://doi.org/10.3389/fonc.2021.618604>.

Feasibility



Dr Andrew Cowburn

*Director of Remote
Medicine at Imperial
College London*



Athanasios Vrontzos

*Expert on future Mars
missions and Machine
Learning-Supported
Diagnostic Systems*



Kenton Kwok

*Former Imperial
Physicist, Master Thesis
focused on ultrasound*

Rapid advancement

Communication delays

Data will be challenging

Hybrid, Human-In-The-Loop System

Impact



Prof Thais Russomano

*Doctor specialising in
space medicine and
space physiology*



Dr Sophie Morse

*Expert on therapeutic
ultrasound, PhD in Non-
invasive Surgery and Biopsy*

Safe and effective

Integrated

Operable



Precision

Reliability

Fail-safe



Multi-functional

Crew autonomy

Standardised



Conclusion



Dr Sophie Morse

*Expert on therapeutic
ultrasound, PhD in Non-
invasive Surgery and Biopsy*

**"I can't think of a better way than doing
things with focused ultrasound in general."**

Feasible
Necessary
Reasonable



Ultrasound modes
Safety
Motion compensation



Q&A

Energy consumption

*efficiency improvements: ultrasound 70% , HIFU 55%, robotic arm 60%, computing systems 90%

Current Technology^[1]

- Ultrasound imaging: 25-35W
- HIFU Treatment Module: 250-350W
- Robotic Arm: 30-40W^[2]
- Computing System:
 - Image Processing: 30-40W
 - AI Diagnostics: 40-70W
- Display/UI: 25-35W

Mode based consumption:

- Active Imaging: 150-180W (**few times weekly**)
- HIFU treatment: 400-500W (rare)
- Standby: 15-20W (most of the time)

- Average Power Consumption
 - 3 imaging sessions/week, 20min each (0.6% of time)
 - 3 HIFU treatments/year (0.01%)
 - Standby (99.4%)

Average Consumption: 18W

Total Energy/year: 158kWh

With tech improvements* in 30 years: 5W

Total Energy/year: 43kWh

[1] K. J. Anneveldt *et al.*, "Waste analysis and energy use estimation during MR-HIFU treatment: first steps towards calculating total environmental impact," *Insights into imaging*, vol. 15, no. 1, Mar. 2024, doi: <https://doi.org/10.1186/s13244-024-01655-2>.

[2] "Discover our Gen3 robots," *Kinova*. <https://www.kinovarobotics.com/product/gen3-robots>

[3] Much, "Much faster progress on energy efficiency is needed to meet global 2030 goal - News - IEA," *IEA*, Nov. 07, 2024. <https://www.iea.org/news/much-faster-progress-on-energy-efficiency-is-needed-to-meet-global-2030-goal>

Feasibility - Research Validation

Robotic HIFU in Autonomous Medical Operations

Technical Feasibility

HIFU is contingent on the accuracy of the scan [X]. Hence, efforts have focused on robotic HIFU for motion compensation and focal control to avoid damaging the surrounding anatomy [X]. NASA Ames Research Centre has developed a robotic surgery testbed [X].

Challenges

The lack of hydrostatic pressure and fluid redistribution due to microgravity will affect acoustic transmission and tissue interaction [X]. Fluid shifts, blood viscosity changes, and microgravity-induced atrophy can also alter the treatment parameters [X].

Considerations

Adaptive coupling is needed for stable ultrasound transmission in microgravity and variable pressure [X]. **Soft robotic actuators** can stabilise patients and restrict tissue motion [X]. **Real-time imaging and radiation-hardened components** will be necessary [X].

AI as a tool for diagnosis and Clinical Decision Support

In 2025, NASA evaluated an AI Clinical Decision Support System (AI CDSS) using OSCE, usually used to assess human clinicians [X]. Preliminary results indicate that it has high accuracy in diagnosis and decision support across various scenarios [X].

The AI CDSS did not perform well in scenarios involving **complex or nuanced cases** that required more contextual understanding [X]. Given that it is a generative AI, it lacks symbolic reasoning critical for plan synthesis and execution [X], and is prone to hallucinations [X].

NASA recognises that it is a rapidly evolving technology and may ultimately be deployable in a zero-tolerance setting [X]. The AI CDSS needs to be **interpretable**, with necessary **guardrails** implemented to ensure alignment with best practices and standard of care [X].

Machine Learning for HIFU Planning, Execution and Real-Time Monitoring

A deep learning technique for HIFU tumour ablation yielded favourable results, with mismanaged and inadequately treated regions accounting for only 0.06% [X]. Manifold learning has been used to detect changes in tissue after HIFU, using low power (30 W) [X].

HIFU uses high frequencies (0.8–3 MHz) for therapy, while diagnostic ultrasound operates at different frequencies (5–15 MHz) [X]. The interference between these two frequency bands can make it tough for an ML model to learn reliable thermal feedback signals [X].

For safety, a **hybrid ML + rule-based system** is preferred [X]. Due to the need for real-time processing of high-resolution data, using **Edge AI** is preferred [X]. Novel methods are being trialed in industry to reduce interference [X].

Feasibility - Expert Evaluation

The system was evaluated by the following experts through interviews. Their opinions have been colour coded for each topic.



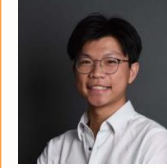
Dr Andrew Cowburn

Director of Remote Medicine at Imperial College London



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Expert on future Mars missions and Machine Learning-Supported Diagnostic Systems



Kenton Kwok

Former Imperial Physicist, Master Thesis focused on ultrasound

Ultrasound technology is advancing rapidly alongside hardware and software, making this system both feasible and necessary for future missions.

- "NVIDIA Jetson being used on the ISS is encouraging. It is a **reasonable assumption** that in 30 years, fault tolerance of components will be of a sufficient competency to be used for Mars."
- "I think it is **feasible** - ultrasound is advancing rapidly at the moment, people are looking at ultrasound in a different way and using it creatively."

Despite advancements, data transmission to Earth would still face round-trip delays, reinforcing the need for the system, alongside onboard processing.

- "As technology advances with robotic intervention, the scanned data can be sent back to Earth for review, but **time delay will still be a problem.**"
- "**The system is definitely not an overkill and will be needed**, from our paper in 2021, telemedicine doesn't work there. 7-15 minutes will be the norm of delay and it can't have real time communication. Therefore, telemedicine can no longer be relied on."

Training the ML model is challenging due to limited data. Insights could come from ISS data or synthetic models.

- "Data is a problem since medical conditions would be different for the general population. Full-tolerance of changing environments will be an issue."
- "It is **possible to access ISS data for an understanding on how the body changes**. The system could be sent to the ISS to get an astronaut to test it over a period of time so that the AI algorithm will learn and understand what's occurring."

It is more likely that a hybrid approach will be taken, such as a Human-In-The-Loop system, built on NASA's safety-critical rule-based framework.

- "Ensure you are building a **rule based system**. This can be a non-ML approach, such as a handbook."
- "Using AI to create a treatment plan is definitely feasible in 30 years, but AI probably won't replace things. **It would be more of a human-AI collaboration.**"
- "I reckon **AI will be involved in 5-10 years**, and it is likely that there will be extensive trials before bringing it to Mars."